DataNova: Automated EDA Report

Format: Executive | Domain Detected: Healthcare

Executive / Dataset Overview - The dataset captures 309 patient records with 16 variables, covering demographics, lifestyle factors, symptom scores, and a binary lung-cancer outcome. - Age ranges from 21 - 87 years, with a median in the mid-50s, reflecting a typical adult patient population for lung-cancer screening programs. - All columns are populated; there are no missing values, enabling immediate model-building or reporting without imputation. - The target variable **LUNG_CANCER** is categorical (positive/negative), making the data ready for classification-focused analytics. - The modest size (≈300 rows) is sufficient for exploratory insights and proof-of-concept models, but larger samples will be needed for production-grade predictive performance.

Data Quality & Profiling Insights - **Consistency:** Every field adheres to its declared data type (e.g., numeric symptom scores, integer age), eliminating the need for datatype cleaning. - **Outliers:** Two age outliers (21 y, 38 y) fall below the statistical lower bound (39 y). They represent a small (<1 %) portion of the cohort and can be retained for completeness or flagged for clinical review. - **Uniformity:** No duplicate rows or contradictory entries were detected, indicating a clean collection process. - **Variable Distribution:** Most symptom and risk factors are recorded on a 0-1 scale, simplifying aggregation and risk scoring. - **Target Balance:** Preliminary glance shows a roughly even split between lung-cancer positive and negative cases, supporting unbiased model training.

Strategic / Operational Takeaways - **Rapid Insight Generation:** With a clean, fully populated dataset, analytics teams can deliver initial risk-factor dashboards and simple predictive prototypes within days, accelerating decision-making for screening resource allocation. - **Targeted Clinical Review:** The two younger age outliers merit a quick chart review to confirm diagnosis accuracy; if validated, they could highlight early-onset cases and inform outreach strategies. - **Scalability Planning:** The current sample size is ideal for pilot studies; however, scaling to a few thousand records will be essential to improve model robustness and to capture rarer sub-populations (e.g., non-smokers). - **Feature Prioritization:** High-impact variables such as **SMOKING**, **YELLOW_FINGERS**, **WHEEZING**, and **SHORTNESS OF BREATH** emerge as immediate candidates for risk-scoring algorithms, enabling quick triage tools for clinicians. - **Compliance & Governance:** Absence of missing data and consistent typing simplify compliance reporting (e.g., HIPAA de-identification audits) and reduce the overhead for data-governance teams.

Visual Analysis